Question				Marking details	Marks Available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
4	*(a)*	(i)		Suitable scales, e.g. 0, 20, 40, 60 on the x-axis and 0, 2, 4, 6 on the y-axis (1) 5 points correctly plotted ± < 1 small square (2) 4 points correctly plotted ± < 1 small square (1) 3 points or less correctly plotted ± < 1 small square (0) Smooth curve of best fit ± < 1 small square (1)	1	2 1		4	4	4
		(ii)		As temperature increases resistance decreases (1) at a decreasing rate (1)		2		2		2
	(b)	(i)		Resistance correct from their graph (1) e.g. 8 [kΩ] $\frac{1}{R}=\frac{1}{8}+\frac{1}{5}$ (1) substitution R = 3.08 [kΩ] (1) $V = IR$ 12 = $I \times 3.08 \times 10^3$ **ecf** on R (1) substitution $I = \frac{12}{3.08\times10^3}$ = 0.0039 [A] (1) **ecf** Answer of 3.9×10^n where n is not -3 award 4 marks **Alternative** [for marks 2 – 5] $I = \frac{V}{R} = \frac{12}{5}$ [= 2.4 mA] (1) substitution $I = \frac{V}{R} = \frac{12}{8}$ = [1.5 mA] (1) substitution Total I = 2.4 + 1.5 = 3.9 [mA] (1) **ecf** addition of currents I = 0.0039 [A] (1)	1 1	1 1 1		5	5	5

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		Resistance of the thermistor increases (1) So current [through the thermistor] decreases (1) But current through the resistor is unchanged (1) So she is (partially) wrong / correct for one but not the other To award full marks the conclusion must be present			3	3		3
				Question 4 total	**3**	**8**	**3**	**14**	**9**	**14**